

A Guide For Building SUSTAINABLE SMART CITIES

Smart cities must fulfil the need for sufficient fresh water, universal access to cleaner energy, ability to travel efficiently from one point to another, sense of safety and security, among others. Smart cities need to be able to efficiently combine a smart physical layer (ability to use sensors to connect physical assets like buildings, public transportation, energy grid, etc) with a smart digital layer (increasingly mixing capabilities like Big Data, AI, the IoT and the like) to crunch data collected and turn it into actionable decisions on the field, in a cost-efficient way



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Smart cities help create an urban environment for adapting to the needs of the population in the most environment-friendly, economically and socially-conscious way. These cities use data from people, vehicles, buildings and things to improve citizens' life and minimise the environmental impact of the cities. These are continuously monitored through the Internet of Things (IoT) to maximise efficiency. Smart cities operate as one big data-driven ecosys-

tem to avoid accidents, emissions and congestion. These employ a combination of low-power sensors, wireless networks, and Web and mobile-based applications, and use technology and data purposefully to make better decisions and deliver a better living experience.

Smart cities are collaborative projects that bring together technology, industry and governments. Technologies used for smart cities should be able to share data in an efficient,

secure and open manner. Successful smart-city projects collect, manage and use data to improve living and workplace standards. Application-driven data is used by both humans and artificial intelligence (AI) to deliver back innovation and efficiencies to the cities.

Why a city needs to be smart

Cities are a pillar of our economic growth. But these can have several inefficiencies with the growing popu-